Mass Conservation and Differential Formulation

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1. From Integral to Differential Descriptions

Integral conservation laws describe the mass, momentum, and energy balances over a finite control volume. To describe a fluid at each point, we apply the same laws to an infinitesimally small region. The result is a set of differential equations relating fluid properties to their spatial and temporal derivatives.

2. The Acceleration Field and Material Derivative

Definition 2.1: Eulerian and Lagrangian descriptions. Eulerian: Specify the velocity at each fixed position and time:

$$\mathbf{V}(x, y, z, t) = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}.$$

Lagrangian: Follow a particular fluid particle with position $\mathbf{x}(t)$. Its velocity is the field evaluated along its trajectory, $\mathbf{V}(\mathbf{x}(t), t)$.

To calculate a particle's acceleration, account for both time dependence and its changing position. The chain rule gives

$$\begin{aligned}\frac{D\mathbf{V}}{Dt} &= \frac{\partial \mathbf{V}}{\partial t} + \frac{dx}{dt}\frac{\partial \mathbf{V}}{\partial x} + \frac{dy}{dt}\frac{\partial \mathbf{V}}{\partial y} + \frac{dz}{dt}\frac{\partial \mathbf{V}}{\partial z} \\ &= \frac{\partial \mathbf{V}}{\partial t} + u\frac{\partial \mathbf{V}}{\partial x} + v\frac{\partial \mathbf{V}}{\partial y} + w\frac{\partial \mathbf{V}}{\partial z}.\end{aligned}$$

Proposition 2.2: Acceleration field.

$$\mathbf{a}(x, y, z, t) = \frac{D\mathbf{V}}{Dt} = \underbrace{\frac{\partial \mathbf{V}}{\partial t}}_{\text{local acceleration}} + \underbrace{(\mathbf{V}\cdot\nabla)\mathbf{V}}_{\text{convective acceleration}}.$$

Local acceleration is the rate at which velocity changes with time at a fixed position. Convective acceleration is the change a particle experiences by moving through a spatially varying velocity field. Both are contributions to its instantaneous acceleration. Hence, a steady flow can still have nonzero particle acceleration.

For any scalar field $f(x, y, z, t)$, such as density, temperature, or pressure, the same chain rule gives

Definition 2.3: Material derivative.

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \mathbf{V} \cdot \nabla f.$$

The material derivative is a particle-following rate of change expressed in Eulerian coordinates. It does not require explicitly solving for a particle's complete trajectory.

Example 2.4: Steady flow can accelerate. In a steady one-dimensional field $u(x) = U(1 + x/L)$, the local acceleration is zero, but the particle acceleration is

$$a_x = u\frac{du}{dx} = \frac{U^2}{L}\left(1 + \frac{x}{L}\right) \neq 0.$$

The flow is steady because u does not depend on time; the particle accelerates because it moves into regions of larger speed.

3. Differential Conservation of Mass

For a fixed control volume, the integral mass balance is

$$\underbrace{\frac{\partial}{\partial t}\int_{\mathrm{CV}} \rho \, dV}_{\text{rate of mass accumulation}} + \underbrace{\oint_{\mathrm{CS}} \rho \mathbf{V} \cdot \mathbf{n} \, dA}_{\text{net outward mass flow rate}} = 0.$$

Consider a small rectangular control volume of dimensions dx, dy, and dz. Only the two faces normal to the x-axis are shown with flow arrows; the y- and z-face contributions follow analogously.

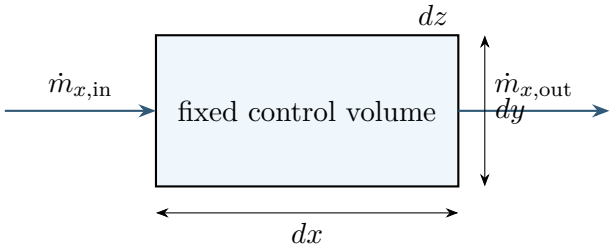


Figure 1: Differential control volume. The mass flux ρu varies slightly between the left and right faces.

If ρ is smooth, the rate of mass accumulation in this small fixed volume is approximately

$$\frac{\partial \rho}{\partial t}\, dx\, dy\, dz.$$

The inlet and outlet mass flow rates in the x-direction are, to first order,

$$\dot{m}_{x,\text{in}} = (\rho u)\, dy\, dz, \qquad \dot{m}_{x,\text{out}} = \left[\rho u + \frac{\partial(\rho u)}{\partial x}\, dx\right] dy\, dz.$$

Hence the net outward rate is

$$\dot{m}_{x,\text{out}} - \dot{m}_{x,\text{in}} = \frac{\partial(\rho u)}{\partial x}\, dx\, dy\, dz.$$

Repeating this for the other four faces, adding accumulation, and dividing by $dx\, dy\, dz$ yields the pointwise equation.

Proposition 3.1: Differential continuity equation.

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0.$$

Equivalently,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0.$$

No assumption of steady or incompressible flow is required.

4. Incompressible Flow

Definition 4.1: Incompressible flow. An incompressible flow is one in which the volume of each material fluid parcel remains constant as it moves. Since a material parcel also conserves its mass, its density remains constant along the trajectory:

$$\frac{D\rho}{Dt} = 0.$$

Expanding the general continuity equation,

$$\frac{\partial \rho}{\partial t} + \mathbf{V} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{V} = 0, \qquad \text{or} \qquad \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{V} = 0.$$

Thus, when $\rho > 0$,

Corollary 4.2: Incompressible continuity equation.

$$\nabla \cdot \mathbf{V} = 0 \quad \Longleftrightarrow \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

Physical meaning. Velocity divergence measures the instantaneous fractional rate at which a material parcel expands or contracts. Zero divergence means its volume is constant, even if its shape changes.

Remark 4.3. Incompressibility does not necessarily imply $\partial\rho/\partial t = 0$. Density can change at a fixed point when different parcels pass through it. Assuming that ρ is constant everywhere is a stronger condition.

5. Physics Olympiad Applications

Given: cylindrical coordinates
For $\mathbf{V} = v_r\mathbf{e}_r + v_\theta\mathbf{e}_\theta + v_z\mathbf{e}_z$,

$$\nabla \cdot \mathbf{V} = \frac{1}{r}\frac{\partial(rv_r)}{\partial r} + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z}.$$

For an axisymmetric flow without swirl,

$$\nabla \cdot \mathbf{V} = \frac{1}{r}\frac{\partial(rv_r)}{\partial r} + \frac{\partial v_z}{\partial z}.$$

Problem 1: Falling water jet, 2020 USAPhO B1, adapted. Water falls steadily from a faucet with downward speed v_0 and jet radius R_0 at the spout. Use (r, θ, z) with z positive upward. Let $R(z)$ be the radius of the stream and let $v(z) > 0$ be the downward speed; the signed vertical velocity component is $v_z = -v(z)$.
Assume the water is incompressible, axisymmetric, and has no swirl. Neglect surface tension and take the dominant vertical speed to be independent of r; the radial component is small compared with the downward velocity. Use the result of part (a) without deriving it:

$$v(z) = \sqrt{-2gz}, \qquad z_0 = -\frac{v_0^2}{2g},$$

where $z = z_0$ at the faucet. Find $R(z)$ without initially assuming $Q = Av$.

Solution. 1. Since $v_z = -v(z)$ and the flow is steady,

$$a_z = v_z\frac{\partial v_z}{\partial z} = v\frac{dv}{dz} = -g.$$

2. Continuity gives

$$\frac{1}{r}\frac{\partial(rv_r)}{\partial r} = \frac{dv}{dz}.$$

Integration in r, requiring finite v_r at the axis, yields

$$v_r(r,z) = \frac{r}{2}\frac{dv}{dz}.$$

3. A surface particle satisfies $r - R(z) = 0$ throughout its motion. Hence

$$\frac{D}{Dt}[r - R(z)] = 0 \quad \Longrightarrow \quad v_r(R(z),z) = -v\frac{dR}{dz}.$$

4. Equating the two boundary expressions gives

$$\frac{R}{2}\frac{dv}{dz} = -v\frac{dR}{dz} \quad \Longrightarrow \quad \frac{d}{dz}(R^2 v) = 0.$$

Using $R(z_0) = R_0$ and $v(z_0) = v_0$,

$$\boxed{R(z) = R_0\sqrt{\frac{v_0}{v(z)}} = R_0\left(\frac{v_0^2}{-2gz}\right)^{1/4}}.$$

The jet narrows as the downward speed increases; the small radial velocity points inward. □

Problem 2: Draining cylinder, 2018 INPhO Q7, adapted. A water-filled cylinder of diameter D is inverted into a basin of water. Air enters through a small hole of diameter d, causing the internal water level $h(t)$ to fall, with $h(0) = h_0 > 0$. Assume constant air and water densities, negligible losses, and $d \ll D$. A pressure and Bernoulli argument gives the air-entry speed

$$v_a(h) = \sqrt{\frac{2\rho_w g h}{\rho_a}},$$

where ρ_w and ρ_a denote the water and air densities.

Solution. 1. The incoming air volume rate equals the decrease in the water-filled volume:

$$\frac{\pi d^2}{4}v_a = -\frac{\pi D^2}{4}\frac{dh}{dt}.$$

2. Substituting the given $v_a(h)$ gives

$$\frac{dh}{dt} = -K\sqrt{h}, \qquad K = \frac{d^2}{D^2}\sqrt{\frac{2\rho_w g}{\rho_a}}.$$

Separating variables and using $h(0) = h_0$,

$$\boxed{h(t) = \left(\sqrt{h_0} - \frac{Kt}{2}\right)^2}, \qquad 0 \le t \le t_{\text{empty}}.$$

3. The modeled emptying time is

$$t_{\text{empty}} = \frac{2\sqrt{h_0}}{K}.$$

The flow is unsteady; both fluid densities are treated as constant in this approximation.

□